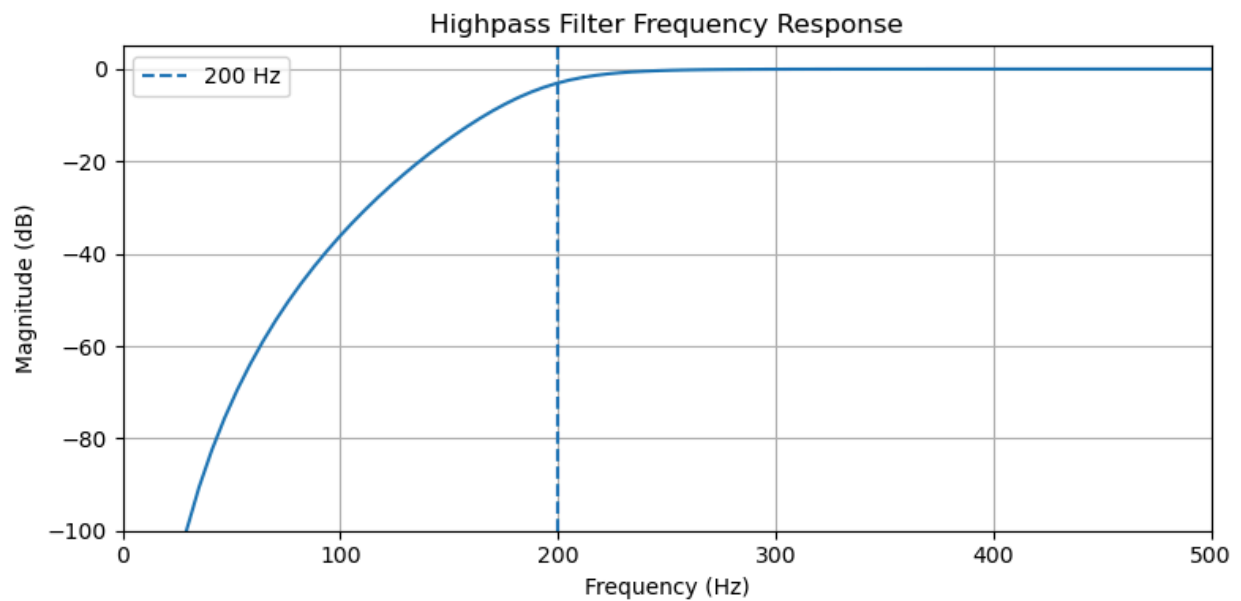
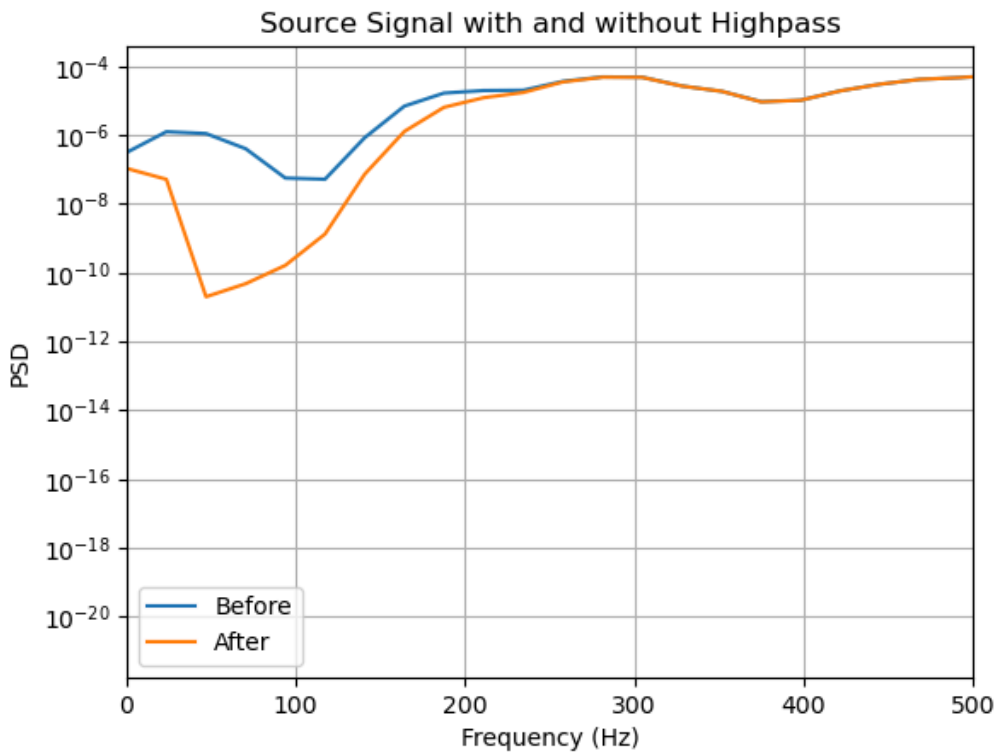
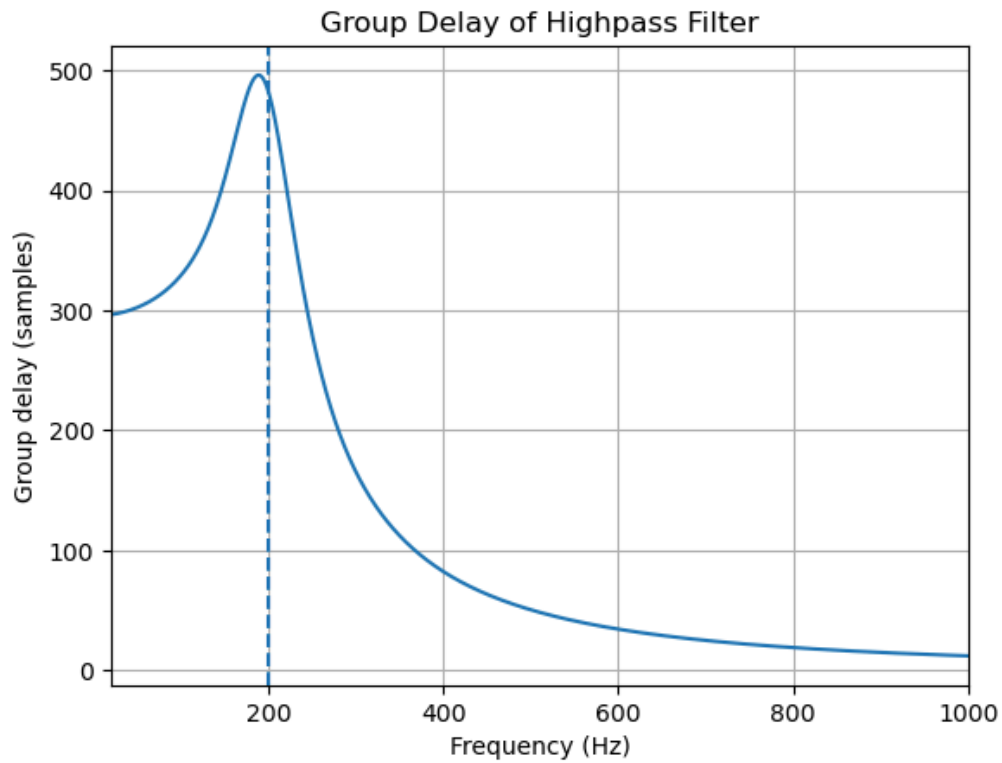


Highpass debugging

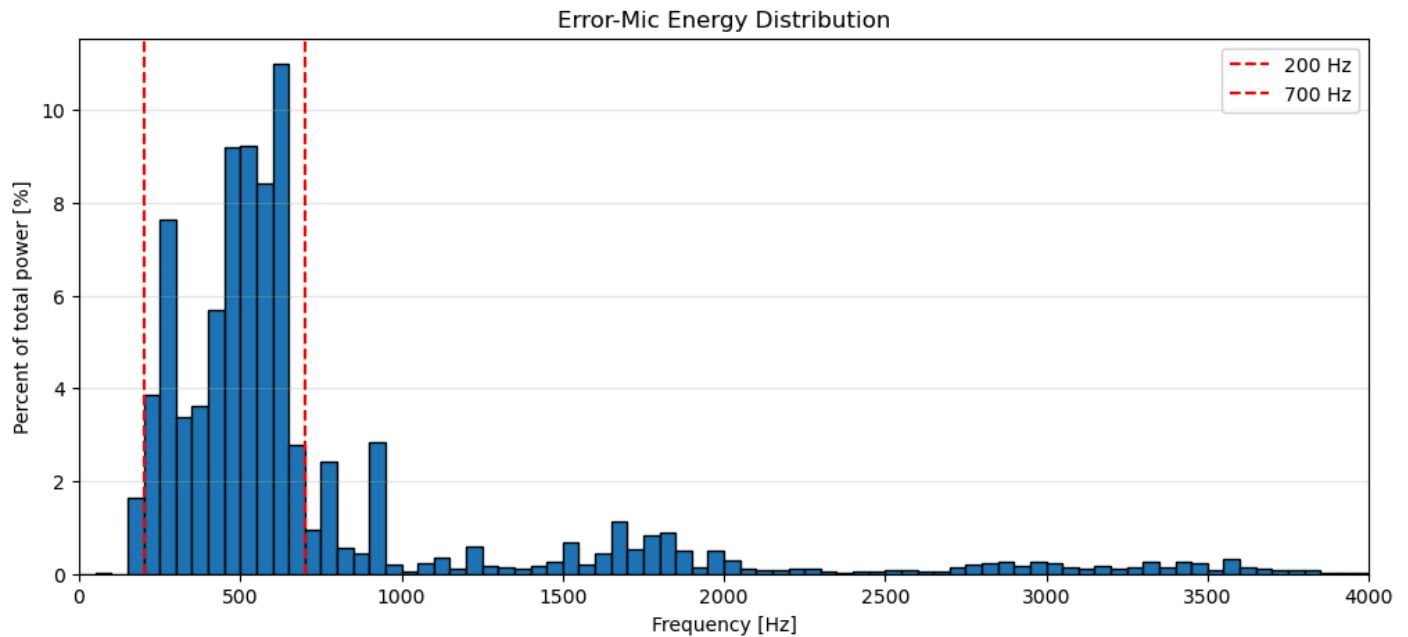
- Highpass seems to correctly remove <200Hz



- Highpass introduces massive group delay near the cutoff

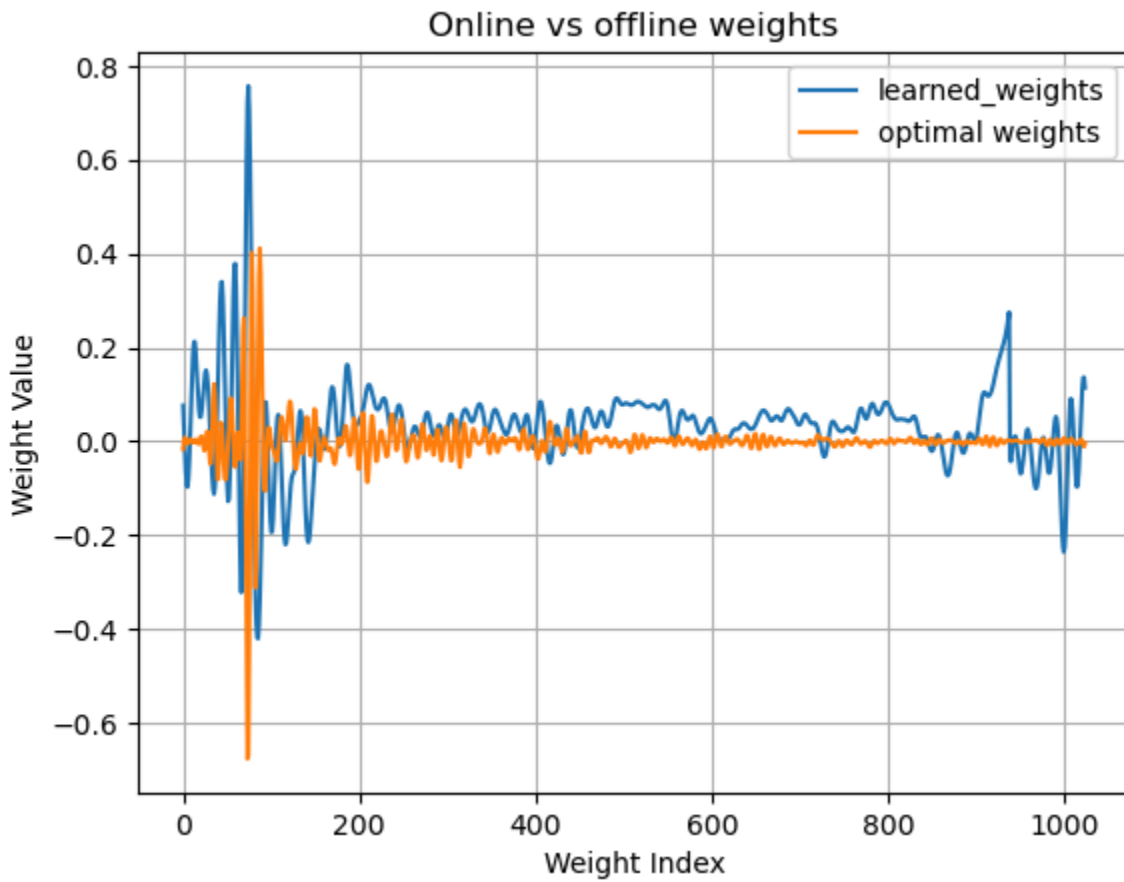


- The <200 Hz is only a tiny portion of the energy of the signal
- I integrated the PSD in each band and divided it by the total area under the PSD

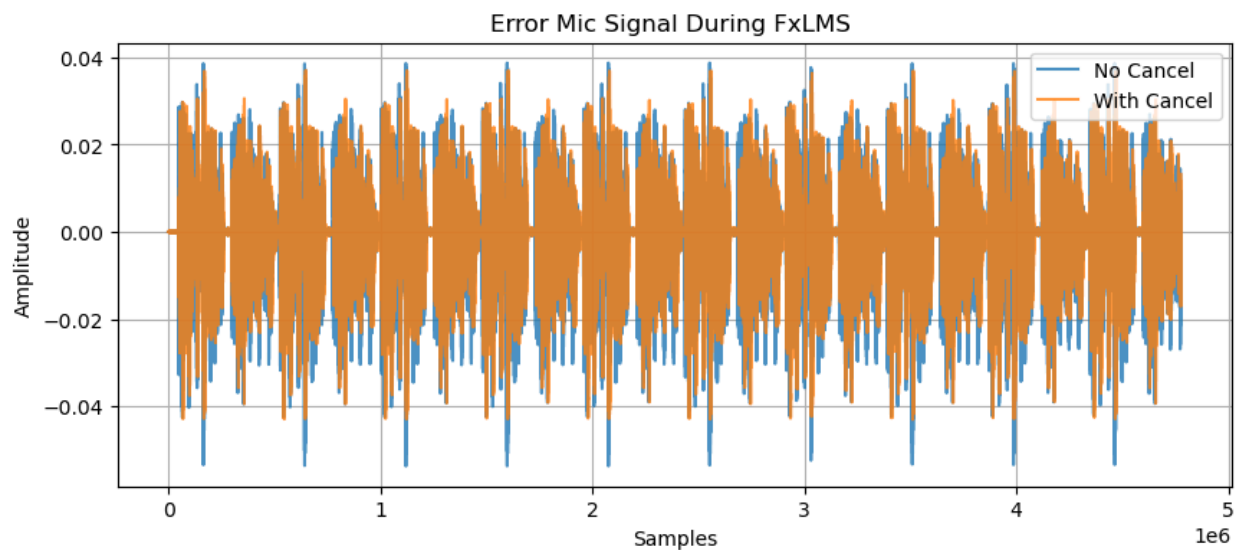


Offline Optimal filter

- I tried seeding the optimal filter learned with built in lag

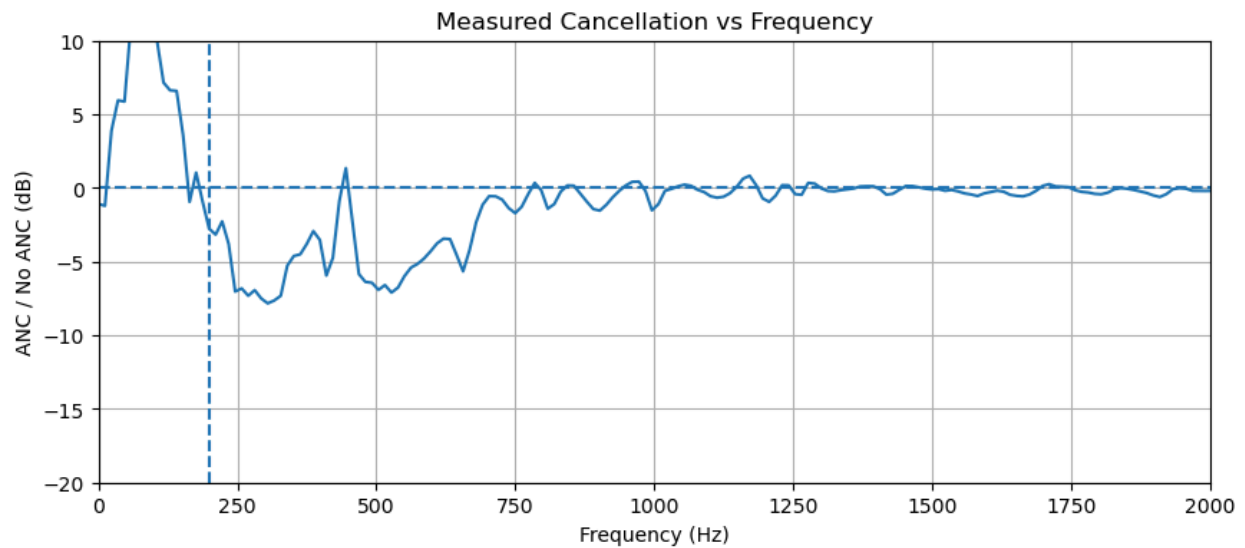


Overall ANC / No ANC: -1.62 dB

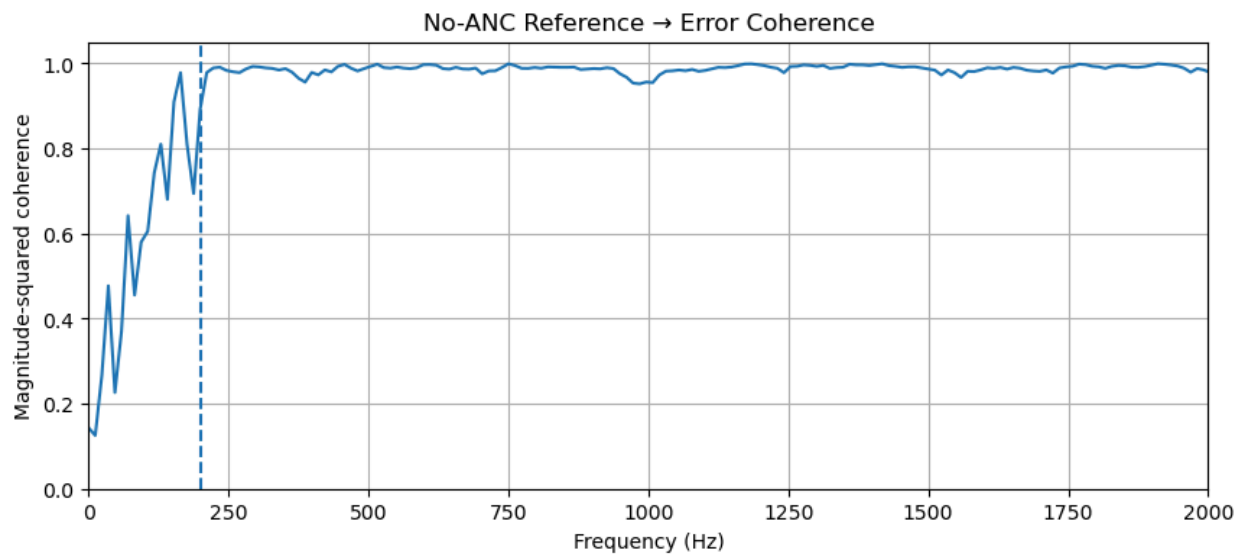


Cancellation Frequency Analysis

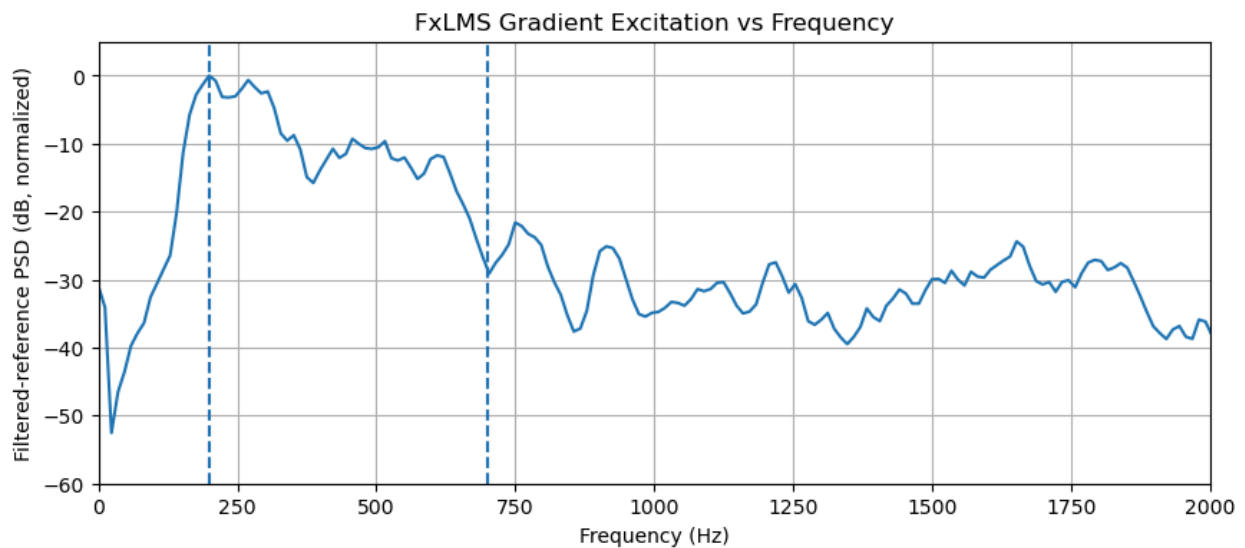
- How does the cancellation work per frequency?



- There is coherence all through the higher frequencies



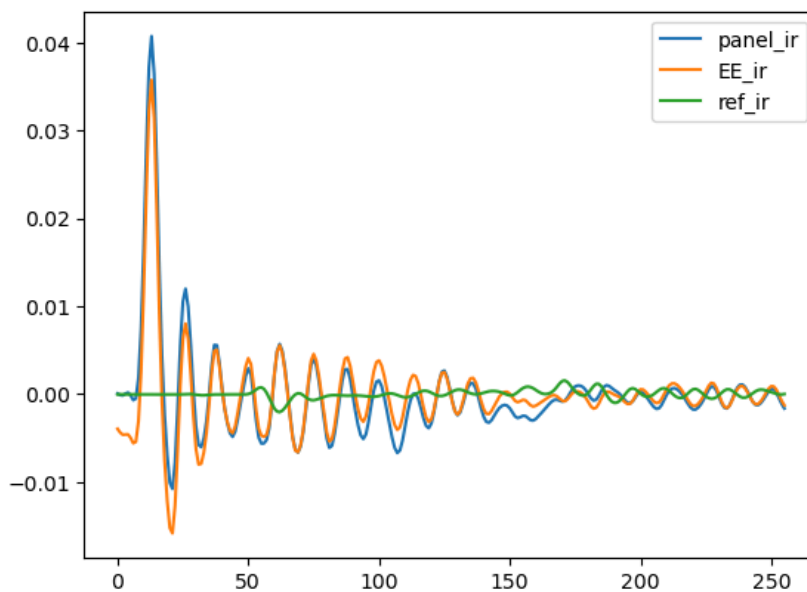
- The PSD of the filtered reference shows that there is a lot more excitation in the main cancel range

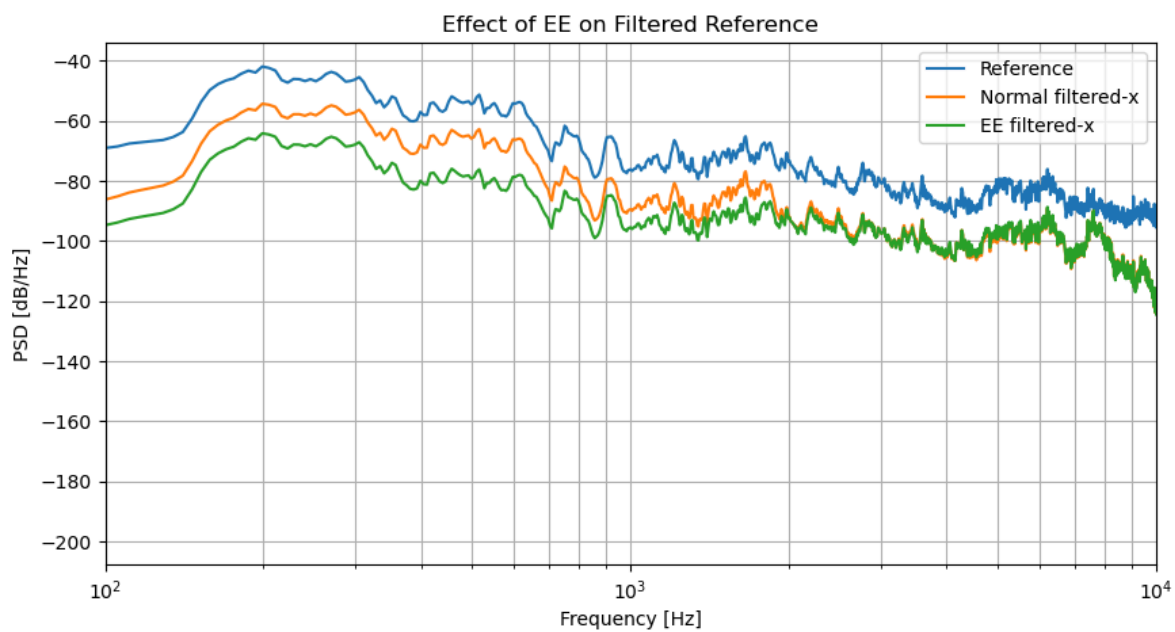
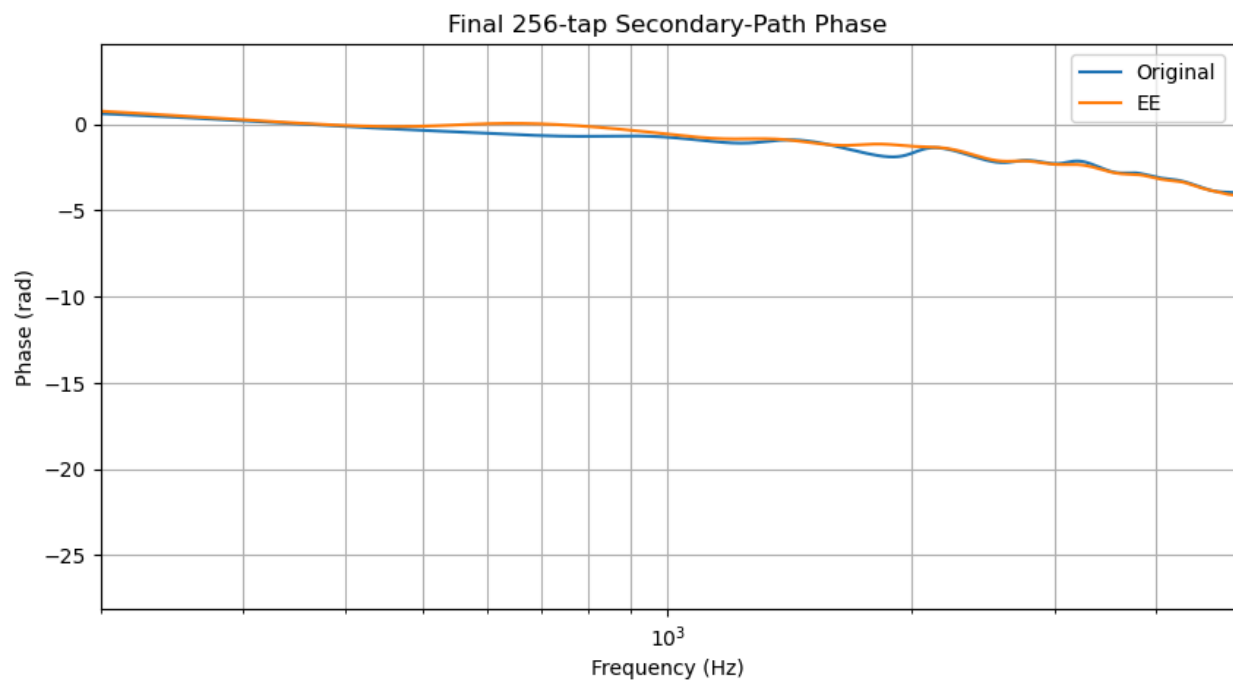


Eigenvalue Equalization

$$|S(\omega)| \propto \frac{1}{|X(\omega)|}$$

- Swap the amplitude of the IR in frequency domain with the inverse magnitude of the reference
- In theory it does what it claims to





- In practice it converges earlier but not better

-1.62 dB

